# Robotic Exoskeleton for Load Transportation

IMU-based locomotion intent recognition and assist gating in MATLAB

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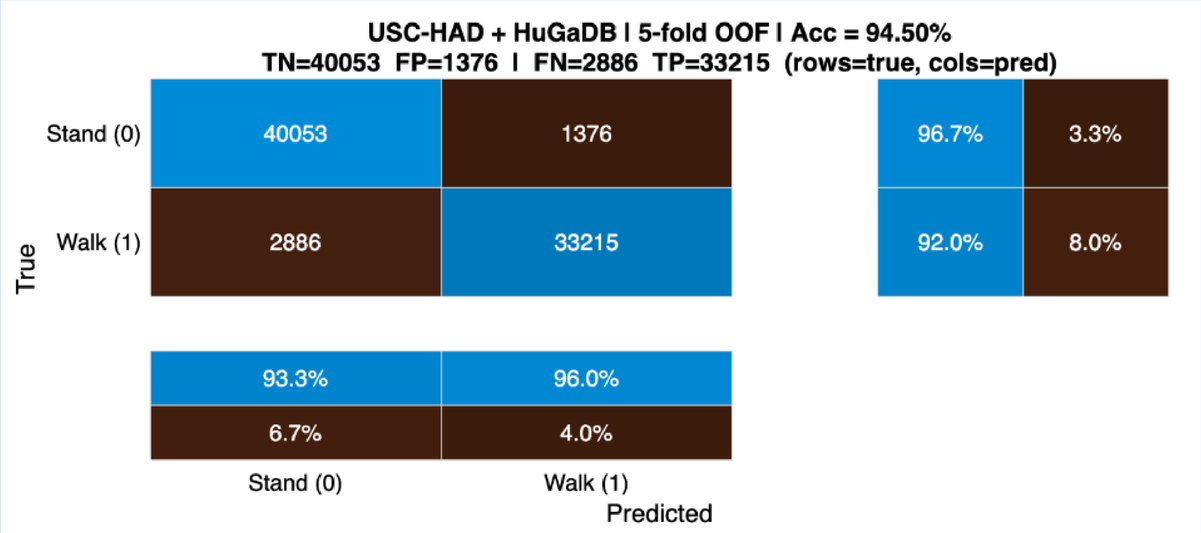
**Key message: a reproducible MATLAB pipeline combines Kalman fusion, binary SVM classification, and FSM hysteresis to turn noisy IMU data into a stable assist on/off command.**

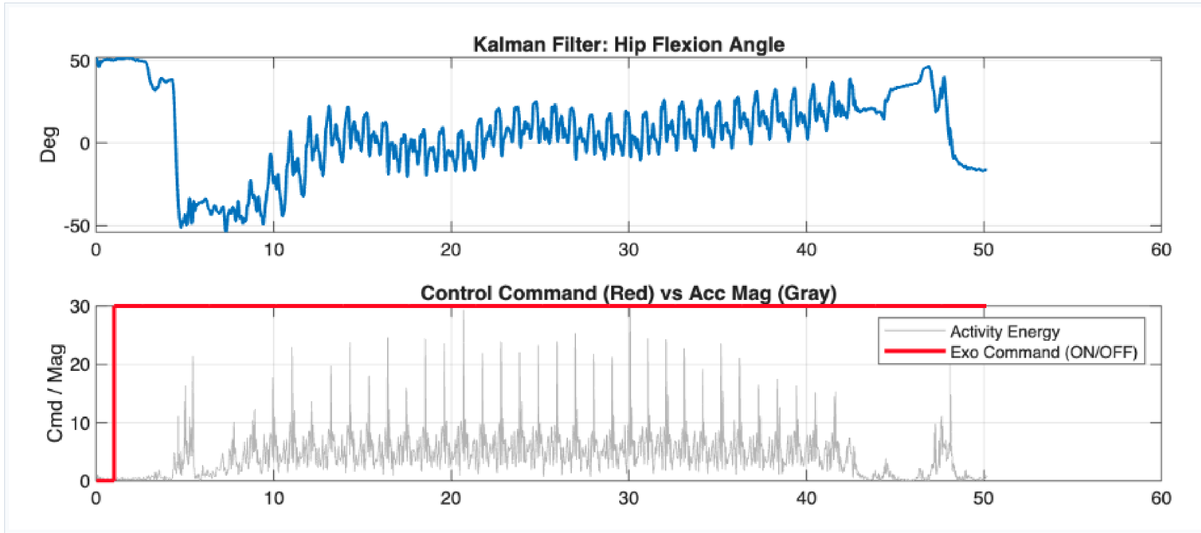
**Deployment baseline**
**Binary SVM, merged 5-fold OOF**
**94.50%**
Best validated claim for assist gating

**Exploratory activity model**
**12-class ECOC, 5-fold OOF**
**56.76%**
Useful for analysis, not direct control

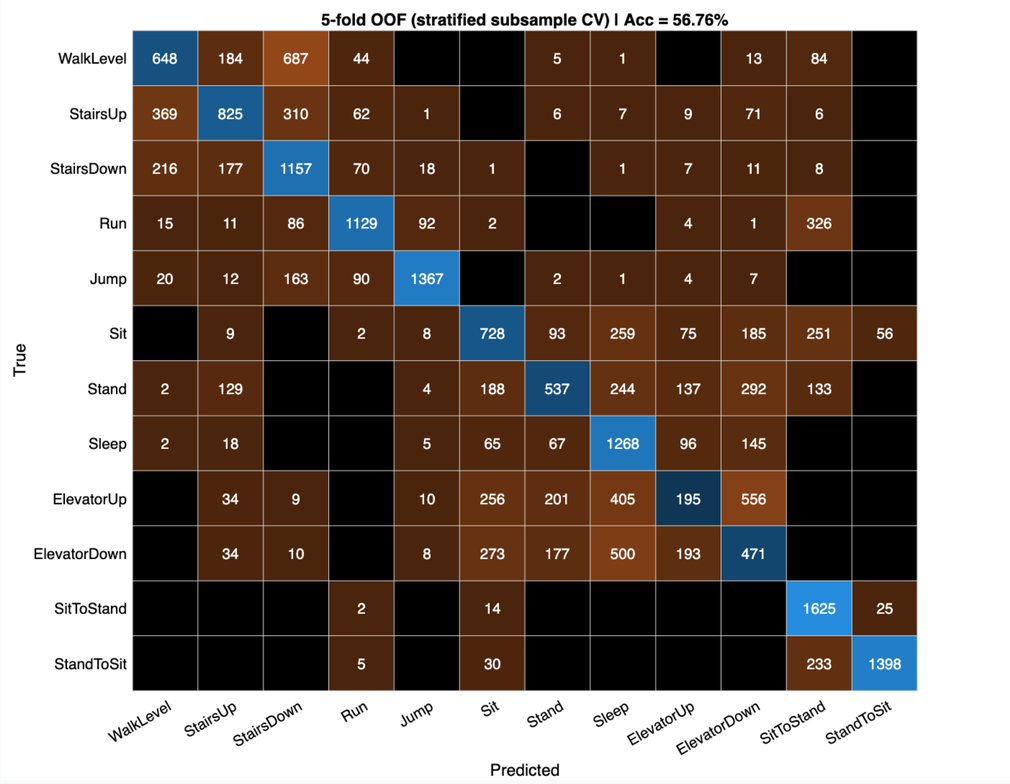
**Sequence baseline**
**Binary LSTM, 20% holdout**
**96.71%**
Separate protocol, not directly comparable



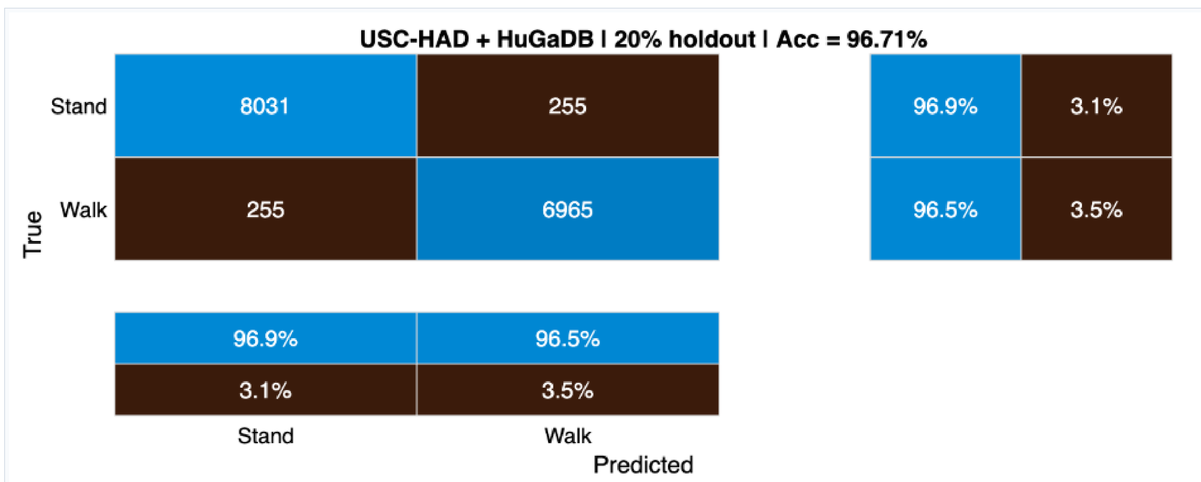

Merged binary SVM: the main validated result for assist gating




Real-time MATLAB replay: fused hip angle and FSM-latched assist output




12-class ECOC: exploratory activity recognition is much harder than binary gating




Binary LSTM: supporting sequence baseline on a separate 20% holdout split

## Pipeline at a glance

1 IMU data and CSV replay → 2 Kalman fusion for hip state → 3 30-D features per window → 4 Binary SVM prediction → 5 FSM hysteresis for assist → 6 Stable on/off command

## Problem and approach

- Manual load handling above 20 kg raises fatigue and musculoskeletal injury risk.
- An exoskeleton is only useful if assistance engages and releases at the right time.
- This work targets a **software supervisor**: infer locomotion-like intent from IMUs and issue a reliable assist on/off signal without EMG.
- The goal is not full torque control yet; it is a robust decision layer that can sit above future hardware control.

## Implemented workflow

- **Input:** public IMU datasets plus a three-phone CSV replay prototype.
- **Fusion:** Kalman-based hip-angle estimation using `imufilter`.
- **Windowing:** 1.0 s windows at 100 Hz with 50% overlap.
- **Features: 30-D** vector, five statistics per IMU slot across six slots.
- **Primary model:** RBF SVM for locomotion-like vs. non-locomotion.
- **Supervisory smoothing:** FSM hysteresis, 3 positives to engage assist and 5 negatives to release.

## Data and evaluation protocol

- **HuGaDB** contributes six on-body IMU streams per window.
- **USC-HAD** contributes one IMU stream; the remaining slots are zero-padded so both datasets share the same 30-D layout.
- The **binary SVM** is the deployment baseline and is evaluated with five-fold out-of-fold predictions.
- The **12-class ECOC** and **binary LSTM** are supporting baselines; the LSTM uses a separate stratified 20% holdout and is therefore not directly comparable.

## Main validated result

The strongest deployable claim is the **merged binary SVM**: it reaches **94.50%** five-fold out-of-fold accuracy while preserving a practical trade-off between false assist activation and missed locomotion onset.

| Training data | OOF acc. |
|---|---|
| USC-HAD only | **98.91%** |
| HuGaDB only | **88.60%** |
| USC-HAD + HuGaDB | **94.50%** |

The USC-only score is numerically highest, but the **merged** model is the better poster result because it generalizes across a broader sensing setup and label mixture.

## How to read the supporting baselines

- The pipeline plot shows why the FSM matters: it turns noisy frame-level predictions into a smoother and more usable assist signal.
- The multiclass ECOC score of **56.76%** is informative for analysis, but not yet strong enough for fine-grained control.
- LSTM holdout accuracy is **96.71%**, but it uses a different validation protocol, so it is a supporting baseline rather than a direct SVM comparison.

## Current limits

- No hardware-in-the-loop or human-subject load trials are reported here.
- The 100–200 ms predictive horizon remains a **design target**, not a re-benchmarked final result.
- Current output is **assist gating**, not full torque scheduling or terrain-aware control.

**Take-home message: the repository now delivers a complete MATLAB baseline for IMU-based binary assist gating, with reproducible public-dataset metrics, CSV replay, sensor fusion, and FSM-stabilized control output.**

## Next steps

- Align SVM and LSTM evaluation under the same cross-validation protocol.
- Add hardware integration, latency measurement, and load-carrying trials.
- Move from binary assist gating toward subject-specific, activity-aware assistance.

**Code and report:**
https://github.com/Rex-Yim/AutomationForExoskeleton

## Summary

- **Delivered:** Kalman fusion, binary SVM, FSM smoothing, reproducible metrics, and CSV replay in MATLAB.
- **Best validated result:** merged binary SVM achieves **94.50%** five-fold OOF accuracy.
- **System role:** simulation-level assist gating supervisor.
- **Value:** a credible, traceable software baseline that separates validated results from future ambition.